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these early documents retain their importance as establishing a definite interest on the part of the Babylonians in arithmetical and geometrical series as early as 700 B. C. and in square and cubic numbers. It is related by Iamblichus and Porphyry that Pythagoras took the harmonical progression from the Babylonians, but these tablets are the only evidence we have of Babylonian interest in series. Proclus, whose historical accuracy is usually not disputed, mentions that the Babylonians were the first to note that six equilateral triangles completely fill the space about a point, but again Babylonian documents to confirm the point are not available.

The contemplation of arithmetical and geometrical series is the most natural and inevitable development of the exercise of the reasoning faculty. Consideration of the number sequence 1, 2, 3, 4, 5, 6, $\dots$ naturally leads to the sequence 1, 3, 5, 7, 9, $\dots$ and such sequences with a known common difference lead more or less naturally, among rational beings, to series in which a common difference is not known but which can be determined by some other condition which is imposed. Among the Greeks and Babylonians the arithmetical series led to the discussion of square and cubic numbers, and to the general subject of number theory, while with the Egyptians the fruit of this contemplation was problems and developments of the kind which we have set forth.

This brief survey of algebraical developments among the Egyptians and Babylonians shows that much of the material which was developed and extended by Greek mathematicians originated, both in methods and in substance, with the scientists of the Orient. The writers of Greece did not hesitate to acknowledge the indebtedness of Greek mathematics to the mathematics of Egypt and Babylon, but nevertheless in recent years real scientific achievements have been denied as emanating from these civilizations. To measure the magnitude of the indebtedness is beyond our power, but to recognize the debt of modern mathematical science to the scientists of Egypt and Babylon is only to render that which is due.

SOME CALCULUS SUGGESTIONS BY A STUDENT.¹

By BENJAMIN GRAHAM, New York City.

Instructors in the calculus are apt to find their own thorough knowledge of their subject somewhat of an obstacle to the complete understanding of their pupils' difficulties. Where long experience has made everything equally clear, it is not easy to feel subjectively the varying degrees of obscurity that enshroud the course as it appears to the eyes of the beginner. Some interest may attach therefore to the following account by a student of his introduction to the calculus,

¹ This paper was prepared several months ago and submitted to the MONTHLY by the author without the knowledge of his instructor. The editors were at once interested and would have been glad to present the paper to the readers of the MONTHLY exactly as it came to them, but, owing to the inexperience of the author in writing for publication, it was found necessary to

including a detailed discussion of what appeared to him at the time to be a serious weakness in the development of the subject, and suggesting a possible remedy for this defect.

Ours was a typical class in a typical large university, with an excellent instructor and Osgood's *Calculus* for a textbook. We sailed quite jauntily through the first semester, finding the theory of differentiation essentially simple and the reverse process of integrating the derivative a natural and even obvious sequel. What difficulty the poorer students experienced came chiefly from an insufficient knowledge of analytical geometry.

It was with the *definite* integral that our troubles started, and it was due to this ingenious contrivance that they never afterwards ended. Instructors are by long familiarity so accustomed to regard the definite integral merely as a new development or division of the calculus that I wonder whether they realize what a logical catastrophe it precipitates in the mind of the inexperienced student.

He has learned that differentiation seeks to discover the rate of change of a given function. Conversely, and in logical sequence, the integral is a function with a given rate of change. Suddenly he is confronted with an entirely new conception. The integral is transformed into the limit of a sum—a notion which, as far as he can see, has absolutely no connection with any previous division of the subject. Connection of some sort is indeed established arbitrarily by the so-called fundamental theorem, which declares the new integral equal to the old. Although it had appeared in general that ours as an academic course aimed rather at rigor of theory than extensive practical application, we were shocked to observe that our textbook waived the proof of this vital proposition. To paraphrase Omar:

"I must abjure the Rate of Change, I must,
Lured by Summation Reckoning, *ta'en on trust*."

Our professor had some inkling of the disastrous consequences this complete change of viewpoint entailed, for he remarked apologetically that we were forced to "swap horses crossing a stream." But I doubt whether he realized fully what hard going we found it on our new mount. In the first place, not only was the concept of a summation entirely foreign to us logically, but its very mechanical expression was cumbersome, unfamiliar, and repellent. Our experience with series had been confined to powers of x , and as we delved more deeply into

revise it in order to put it into suitable form. After submitting it to two or three well-known professors of mathematics, all of whom recognized that this young man had hit upon a weak spot in the teaching of the calculus, but none of whom seemed willing to undertake a revision of the paper, the editors, by mere chance, referred it to the very instructor in question. He has taken it up with the author in personal conferences and by his assistance the author has been enabled to rewrite the paper in its present form.

Strangely enough, two or three other papers have, during this time, been presented to the MONTHLY proposing substitutes for, or modifications of, Duhamel's theorem. From these the editors have selected the one by Professor Huntington, printed in this issue, as representing the best treatment. Doubtless many will find "food for thought" in these papers which may lead to contributions in the Department of Discussions.

EDITORS.

problems involving the definite integral we became lost in a maze of letters and subscripts.

This might be called a small matter, were it not that our awkwardness in handling summations produced one most unfortunate result, namely, that from the very start we failed to understand thoroughly the *necessity* for Duhamel's Theorem. That is to say, most of us were unable to tell accurately which expressions could be summed directly or which needed an intermediary process to aid the integration. Of course, with this handicap to begin with, it was a foregone conclusion that the whole elaborate mechanism of Duhamel's Theorem should ultimately prove an absolute mystery to most of us, and should be seen only in a glass darkly by our brightest stars.

As may be imagined, these basic difficulties did not diminish with the advent of double and triple integrals. The net result of our cumulative bewilderments is easy to summarize. As for scientific rigor of theory, we had it not. The rank and file of the class solved problems as best they could by securing an approximate expression for an element and integrating it by the aid of Pierce's Tables. We knew the answer was correct and perforce were content. Of that logical discipline which the Calculus is so eminently fitted to impart and which alone perhaps may justify its inclusion in a purely academic curriculum, there remained scarcely a vestige.

This sad tale is entirely autobiographical. The writer feels entitled to speak authoritatively, not only because he learned of these mental reactions at first hand from his fellow students, but also because he experienced them largely himself. It was only by additional efforts that he finally mastered the theory of the definite integral—efforts inspired by a special interest in the subject and which the student cannot usually be expected to make. As his own original difficulties and those of his class remained very vividly in his mind, he was led to devise an entirely different method of attacking problems in the integral calculus, which he believed would simplify both the theory and the technique of solving problems by integration.

The process suggested by Professor Bliss in the *ANNALS OF MATHEMATICS* in 1914 obviates to a great extent the technical complexity of Duhamel's Theorem. But aside from the mechanical difficulties of the summation method, the writer feels that if the original rate-of-change conception were retained throughout the calculus the student would gain a more consistent and unified idea of the meaning of the subject as a whole.

The solution of problems by use of the indefinite integral consists in finding an expression for the derivative—or rate of change—of the desired quantity, and then integrating this expression. The application of this method is limited by the fact that in most cases the derivative is not expressible as some function of the unknown variable, and hence is not in integrable form. The analogous difficulty of the definite integral, or summation, process—namely that the elements cannot be summed directly—is surmounted by such artifices as Duhamel's Theorem. The ensuing propositions aim to perform a like service for indefinite integrals.

All functions discussed are assumed to be continuous over the interval considered and therefore have maximum and minimum values in each of such intervals.

We will assume the following definitions in which $y = f(x)$:

$$(I) \quad dy = \left(\lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} \right) \cdot dx, \quad (II) \quad y = \int \left(\lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} \right) \cdot dx.$$

THEOREM 1. If $x_k < x_k' < x_k + \Delta x$, then $\lim_{\Delta x \rightarrow 0} f(x_k') = f(x_k)$.

Proof. Let $x_k' = x_k + \Delta'x$. Then $x_k < x_k + \Delta'x < x_k + \Delta x$ by hypothesis and $0 < \Delta'x < \Delta x$. When $\Delta x \rightarrow 0$, $|\Delta x|$ can be made less than any assignable quantity, such as ϵ . But when $|\Delta x| < \epsilon$, then $|\Delta'x| < \epsilon$ since $\Delta'x$ is positive and $< \Delta x$. Therefore, by definition of limit, $\lim_{\Delta x \rightarrow 0} \Delta'x = 0$, $\lim_{\Delta x \rightarrow 0} x_k' = x_k$, $\lim_{\Delta x \rightarrow 0} f(x_k') = f(x_k)$ by definition of continuous function.

THEOREM 2. If $\Delta Q = f(x_k') \cdot \phi(x_k'') \cdots \Delta x$, where $x_k < x_k', x_k'' \cdots < x_k + \Delta x$, then $Q = \int f(x) \cdot \phi(x) \cdots dx$.

$$\begin{aligned} \text{Proof. } Q &= \int \lim_{\Delta x \rightarrow 0} \frac{\Delta Q}{\Delta x} \cdot dx && \text{Definition (II)} \\ &= \int \lim_{\Delta x \rightarrow 0} \frac{f(x_k') \cdot \phi(x_k'') \cdots \Delta x}{\Delta x} \cdot dx = \int \lim_{\Delta x \rightarrow 0} f(x_k') \cdot \phi(x_k'') \cdots dx \\ &= \int f(x_k) \cdot \phi(x_k) \cdots dx && \text{by Theorem 1.} \end{aligned}$$

Dropping subscripts,

$$Q = \int f(x) \phi(x) \cdots dx.$$

THEOREM 3. LEMMA.—A function, $f(x)$, continuous in the interval (a, b) takes at least once every value comprised between $f(a)$ and $f(b)$ for a value of x comprised between a and b .

This Theorem is proved in Goursat's *Cours d'Analyse Mathématique*, pp. 162–3, and we shall therefore omit the demonstration here.

A special case of Theorem 2 will be found useful in practice.

THEOREM 2a. If $\Delta Q = f(x_k') \cdot \phi(x_k'') \cdots \Delta x$ and $f(x_k''') \leq f(x_k') \leq f(x_k^{IV})$, $\phi(x_k^V) \leq \phi(x_k'') \leq \phi(x_k^{VI}) \cdots$, where $f(x_k''')$ and $\phi(x_k^V)$ are minimum values, and $f(x_k^{IV})$ and $\phi(x_k^{VI})$ are maximum values of these functions over the continuous interval $x_k < x < x_k + \Delta x$, then

$$Q = \int f(x) \cdot \phi(x) \cdots dx.$$

Proof. $f(x_k''') \leq f(x_k') \leq f(x_k^{IV})$. Then for some value of x_k' in our interval,

$$x_k''' \leq x_k' \leq x_k^{IV} \quad \text{by Theorem 3.}$$

But by hypothesis,

$$x_k \leq x_k''', \quad x_k \leq x_k^{IV} + \Delta x.$$

$$\therefore x_k \leq x_k' \leq x_k + \Delta x.$$

Similarly

$$x_k \leq x_k'' \leq x_k + \Delta x.$$

Then by Theorem 2,

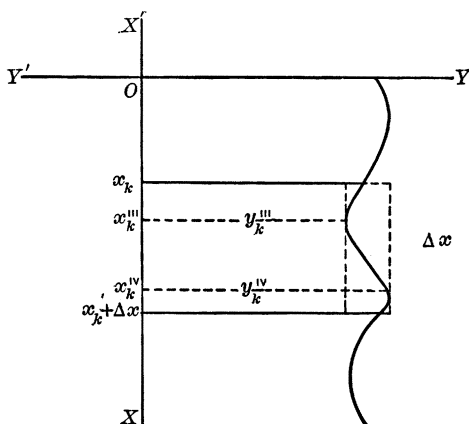
$$Q = \int f(x)\phi(x) \cdots dx.$$

By means of Theorems 2, or 2a, and 3 the problems usually treated by the summation process and Duhamel's Theorem may be readily solved in rigorous fashion by the indefinite integral, *i. e.*, by finding the derivative of the required function and integrating it. The application of our method to a problem involving simple integrals will serve to make it clearer.

We have selected the water-pressure example, since the solution here given can readily be compared with the current method as found, say, in Osgood's *Calculus*, and with the simplified solution worked out by Professor Bliss in the above-mentioned article.

Problem 1. To find the pressure of a liquid on a vertical wall.

Solution. Let the wall be bounded as in the figure, the Y axis lying in the surface of the liquid. At the point x_k , let the abscissa suffer an increment Δx . The area A_k and the pressure P_k will take on corresponding increments ΔA and ΔP .



By physics,

$$\Delta P = wx_k' \Delta A, \quad (1)$$

where w is the weight per unit of liquid and where

$$x_k < x_k' < x_k + \Delta x.$$

From the figure $y_k''' \Delta x < \Delta A < y_k^{IV} \Delta x$, where y_k''' and y_k^{IV} are respectively maximum and minimum ordinates of ΔA .

Let

$$\Delta A = y_k'' \Delta x.$$

Then

$$y_k''' < y_k'' < y_k^{IV}, \quad (2)$$

$$\Delta P = wx_k' y_k'' \Delta x, \quad (3)$$

whence applying Theorem 2a to (1), (2) and (3),

$$P = \int wxy dx.$$

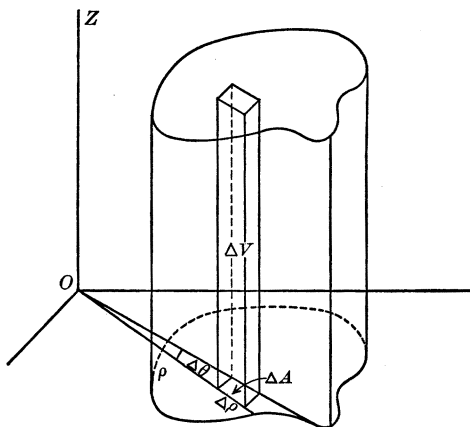
By a simple process of generalization our theorems can be extended to apply to integrals of higher order. To save space we will dispense with the demonstrations and confine ourselves to an example of their application to a problem involving a double integral.

Problem 2. To find the volume under a surface in cylindrical coördinates.

Solution. Let the equation of the surface be

$$z = f(\rho, \theta).$$

To the angle $\theta = \theta_k$ add an increment $\Delta\theta$. To the radius vector $\rho = \rho_k$ add an increment $\Delta\rho$. On the small area ΔA thus formed construct the increment of volume ΔV , in the form of a right cylinder but terminated by the given surface.



Let

$$\Delta V = z_k' \Delta A.$$

By geometry,

$$\Delta A = \frac{\Delta\theta}{2} (\rho + \Delta\rho)^2 - \frac{\Delta\theta}{2} \rho^2,$$

whence

$$\Delta A = \left(\rho + \frac{\Delta\rho}{2} \right) \Delta\theta \Delta\rho,$$

$$\rho < \rho + \frac{\Delta\rho}{2} < \rho + \Delta\rho. \quad (1)$$

From the figure, $z_k''\Delta A < \Delta V < z_k'''\Delta A$, where z_k''' and z_k'' are respectively maximum and minimum values of z in ΔV .

Then

$$z_k'' < z_k' < z_k''', \quad (2)$$

$$\Delta V = z_k' \left(\rho + \frac{\Delta \rho}{2} \right) \Delta \theta \Delta \rho, \quad (3)$$

whence, applying the generalized form of Theorem 2a to (1), (2) and (3),

$$V = \int d\rho \int z \rho d\theta.$$

The writer believes that the foregoing method could be used throughout the calculus, thus conserving the conception of the rate of change of a function as the basis of the entire subject. Most instructors would probably object to the elimination of the summation process on the ground that the latter is the most natural and vivid method of attacking practical problems. But is it not possible that the summation idea appears most natural to them only because they have been so long accustomed to it? To the beginner there is not very much logical clearness about the sudden jump from the inverted derivative notion to that of the integral as the limit of a sum. Certainly the former method presents no logical difficulties, and its retention throughout the calculus would at least add unity and consistency to the course.

The engineer is supposed to solve all his problems by the summation process. But after all, his method consists merely of setting an integral sign before the approximate value of a single element. Call this element an increment, and his integral reduces to the simple anti-derivative. For this intuitive process, the foregoing theorems would seem to provide a rigorous mathematical foundation.

ON SETTING UP A DEFINITE INTEGRAL WITHOUT THE USE OF DUHAMEL'S THEOREM.¹

By EDWARD V. HUNTINGTON, Harvard University.

The purpose of this note is to state a simple theorem by means of which the ordinary process of "setting up an integral" may be simplified and made entirely rigorous without the use of Duhamel's Theorem or any of its modern substitutes. In view of the fundamental importance of the process of setting up an integral, it is hoped that such a simplification may be of value in both pure and applied mathematics.

To fix our ideas, let us take the familiar problem of finding the total attrac-

¹ This note contains the substance of two papers presented to the American Mathematical Society, April 29 and September 5, 1916, under the titles: (1) A simple example of the failure of Duhamel's Theorem, and (2) A simple substitute for Duhamel's Theorem.